

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			<u>Protein</u>	1			1		
	(b)			Increases then decreases (1) optimum pH/ pH 8 (1) Award two marks for: peaks at {optimum pH/pH 8}		2		2		
	(c)	(i)		Ref to {changed shape/destruction of} (enzyme/active site)		1		1		
		(ii)		- nothing for the <u>substrate</u> (to fit into)/ cannot make enzyme <u>substrate</u> complexes (1) - No reaction/ will not react (1) ignore no products		2		2		
	(d)			pH/ acid/ alkali (1) (high) temperature/boiling (1) Ignore heat	2			2		
				Question 3 total	3	5	0	8	0	0